Geoffrey Mesbahi

geoffrey.mesbahi@gmail.com
geoffreymesbahi.netlify.app
ORCID: 0000-0003-0276-3901

Research experience

Researcher
Since 2022

Grassland management and ruminant nutrition

Main projects:
- SCALE-it WP4 and WP5 (Horizon 2020)
- ReLivestock WP2 (Horizon 2020)
- Charclim (Canton Aargau)
- Weidelabor (Canton Aargau and Migros)

Skills:
- Forage sampling
- Faeces and rumen fluid sampling
- Feeding behaviour sampling (scan and focal sampling)
- Agroforestry plot design
- Tree planting and pruning
- Statistical analysis (R: mixed models, machine learning)
- GIS (QGIS)
- Team management
- Supervision of 3 MA
- Outreach

Post-doc
2020-2022

Agronomic value of temperate fodder trees for dairy cattle production
PhD Sandra Novak, INRAE, France

- Forage sampling
- Faeces sampling
- Feeding behaviour sampling (scan and focal sampling)
- Remote sensing (drone photogrammetry, TLS)
- Statistical analyses (R: mixed models, model selection, HCPC)
- GIS (QGIS)
- Supervision of 1 MA, 2 BA students, and 1 field assistant
- Pollard trees maintenance
- Outreach

PhD
2017-2020

Prediction of agronomical and ecological grasslands characteristics, and their trade-offs. Focus on the Vosges Mountains permanent grasslands
Prof Sylvain Plantureux, University of Lorraine - INRAE, France
Cécile Bayeur, Northern Vosges Regional Nature Park, France

- Field botany
- Forage sampling
- Soil sampling
- Interviews of farmers
- Statistical analyses (R: model selection, random forest, HCPC)
- GIS (QGIS)
- Supervision of 7 MA students
- Teaching 43 hours
- Outreach

Field assistant
2015-2016

Semantics content in titi monkey alarm call sequences
PhD Mélissa Berthet, University of Neuchâtel, Switzerland

- Wild monkey population monitoring
- Cognitive experiments
- Behaviour observation (scan sampling)
- Faeces sampling
- Protocol development
- Field assistants training

Engineer thesis
2014

Impact of soil properties and agronomical practices, and their interactions, on French earthworms communities
PhD Kevin Hoeffner, University of Rennes 1 - CNRS, France

- Earthworms gathering
- Earthworms identification
- Database management
- Statistical analyses

Education

PhD
2017-2020

PhD in agronomic sciences
University of Lorraine Lorraine, France
Jury: Prof Gillet F., PhD Bonis A., PhD Diquelou S., PhD Magda D., PhD Michaud A.

Engineer degree
2009-2014

Engineer in agriculture and management
UniLaSalle Beauvais, France

Additional courses

Workshop
2022

Arkesys, France
Bayesian statistics with R: initiation and training

Summer school 2021	**Analysis and Experimentation on Ecosystems**, France Remote sensing for the study of terrestrial ecosystems: From theory to practical application
Animal testing 2021	**ONIRIS**, France Animal testing diploma: experimental design and execution
Workshop 2021	**Gaia formation**, France Fodder trees for herbivorous farm animals
Workshop 2019	**Regional Nature Parks and Botanic conservatories**, France Grassland phytosociological classification
short training courses	**University of Lorraine**, France Writing and publishing a paper, Pedology, Life cycle assessment
	INRAE, France Writing a data paper, Choosing a scientific journal, HAL, DataINRAE

Grants

AnaEE 2021	Scholarship (1 week) to participate in the AnaEE summer school on remote sensing at the Lautaret Garden, France.
INRAE Phase 2021-2022	"Projet Innovant" Fund (19 000€, 2 years) for the project "*Agronomic value of temperate fodder trees for dairy cattle production*"
INRAE AgroEcoSystem 2021-2022	"Pari Scientifique" Fund (8 250€, 2 years) for the project "*Agronomic value of temperate fodder trees for dairy cattle production*"
Fondation de France 2021-2022	Postdoctoral grant (93 370€, 2 years) for my postdoc "*Agronomic value of temperate fodder trees for dairy cattle production*"
Agence de l'eau Rhin-Meuse 2020	Funding (22 500€, 8 months) for the completion of my PhD thesis "*Prediction of agronomical and ecological grasslands characteristics, and their trade-offs. Focus on the Vosges Mountains permanent grasslands*" and the redaction of the book "*Les prairies permanentes du massif des Vosges. Typologie agroécologique et diagnostic prairial*"

Journal articles

Agronomy, ecology

Mesbahi G., Leiber F., Stein A. K., Leubin M., Thüer S., Dittmann M. T., Vitra A., Werner J. (2026) The methanogenic potential of grass-fed dairy calves is mostly impacted by the feeding system, not the genotype. *Livestock Science*, under review

Leiber F., Baki C., Bieber A., Mesbahi G., Werner J., Spengler Neff A. (2025). Grazing enhances immune parameters and complete blood count in calves. *Swiss Agricultural Research, 16, 206-211*
doi: 10.34776/afs16-206e

Mesbahi G., Barre P., Delagarde R., Ginane C., Novak S. (2025) Seasonal dynamics of the nutritive value of temperate forage trees differ among species. *Agroforestry Systems*, 99, 172
doi: 10.1007/s10457-025-01258-w

Galland S., Dind A., Schmid N., Mesbahi G., Dubois S., Probo M., Mariotte P. (2025) Rendement et valeur nutritive de sept espèces d'arbres fourragers. *Recherche Agronomique Suisse, 16, 66-72*
doi: 10.34776/afs16-66

Trouillard M., Bosshardt S., Derbez F., Desaint B., Dufils A., Mesbahi G. (2024) Agroforestry: benefits and considerations to meet the challenges of organic livestock production. *INRAE Productions Animales, 37 (2), 7468*
doi: 10.20870/productions-animales.2024.37.2.7468

Berthet M., Mesbahi G., Duvot G., Zuberbühler K., Cäsar C., Bicca-Marques J.C. (2021) Dramatic decline in a titi monkey population after the 2016-2018 sylvatic yellow fever outbreak in Brazil, *American Journal of Primatology*, e23335
doi: 10.1002/ajp.23335

Mesbahi G., Michelot-Antalik A., Goulnik J., Plantureux S. (2020) Permanent grassland classifications predict agronomic and environmental characteristics well, but not ecological characteristics, *Ecological Indicators*, 110, 105956
doi: 10.1016/j.ecolind.2019.105956

Goulnik J., Plantureux S., Van Reeth C., Baude M., Mesbahi G., Michelot-Antalik A. (2020) Facial area and hairiness of pollinators visiting semi-natural grassland wild plants predict their facial pollen load, *Ecological Entomology*, 45 (6), 1296-1306
doi: 10.1111/een.12913

Mesbahi G., Bayeur C., Michelot-Antalik A., Plantureux S. (2019) Choosing classification systems for predicting the characteristics of permanent grasslands, *Fourrages*, 237, 57-65

Animal communication

Berthet M., Mesbahi G., Leroux M. (2026) The As and Bs of titi monkey linguistics: why emotional communication is not the enemy, *Biological Reviews*
doi: 10.1002/brv.70131

Berthet M., Mesbahi G., Cäsar C., Zuberbühler K. (2022) Impact of predator model presentation paradigms on titi monkey alarm sequences, *Behavioral Ecology and Sociobiology*, 76
doi: 10.1007/s00265-022-03250-1

Narbona Sabaté L., Mesbahi G., Dezecache G., Cäsar C., Zuberbühler K., Berthet M. (2022) Animal linguistics in the making: The Urgency Principle and titi monkeys' alarm system, *Ethology Ecology & Evolution*
doi: 10.1080/03949370.2021.2015452

Berthet M., Mesbahi G., Pajot A., Cäsar C., Neumann C., Zuberbühler K. (2019) Titi monkeys combine alarm calls to create probabilistic meaning, *Science Advances*, 5:eaav3991
doi: 10.1126/sciadv.aav3991

Berthet M., Neumann C., Mesbahi G., Cäsar C., Zuberbühler K. (2018) Contextual encoding in titi monkey alarm call sequences, *Behavioral Ecology and Sociobiology*, 72:8 doi: 10.1007/s00265-017-2424-z

Book & Thesis

Bayeur C., Hennequin C., Mesbahi G., Plantureux S. (2020) Les prairies permanentes du massif des Vosges. Typologie agroécologique et diagnostic prairial.
ISBN: 978-2-9574260-0-3

Mesbahi G. (2020) Prediction of agronomical and ecological grasslands characteristics, and their trade-offs. Focus on the Vosges Mountains permanent grasslands (PhD thesis).

Conferences and posters

Agronomy, ecology

Werner J., Leiber F., Vitra A., Mesbahi G. (2025) Performance of veal calves of different genotypes raised on contrasted multi-species grasslands. *23rd Symposium of the European Grassland* Federation, Reading, United Kingdom

Poster and conference paper

Mesbahi G., Steiner A. K., Leiber F., Werner J. (2025) Farming system is the main driver of enteric methanogenesis in grass-based veal calves. *76th Annual Meeting of the European Federation of Animal Science*, Innsbruck, Austria

Oral presentation and conference paper

Werner J., Leiber F., Mesbahi G. (2025) Performance of male dairy calves of different breeds raised on various types of grasslands for veal production. *45th Swiss Animal Nutrition Conference*, Lindau, Switzerland

Oral presentation

Mesbahi G., Steiner A. K., Leiber F., Werner J. (2025) Enteric methane production in calves depends primarily on the farming system, not on breed. *45th Swiss Animal Nutrition Conference*, Lindau, Switzerland

Poster presentation

Kadiri Y., Mesbahi G., Delagarde R., Barotin C., Novak S. (2024) Composition chimique et digestibilité enzymatique des tiges et des feuilles de 4 espèces. *27e Rencontres autour des recherches sur les ruminants*, Paris, France

Poster presentation and conference paper

Mesbahi G., Vitra A., Dittmann M. T., Steiner A. K., Thorne S., Hesselmann M., Leiber F. (2024) Trade-offs between milk production potential, tannin content and plant diversity depend on grassland mixtures. *30th General Meeting of the European Grassland Federation*, Leeuwarden, The Netherlands

Poster presentation and conference paper

Maxin G., Graulet B., Novak S., Mesbahi G., Signoret F., Glinec J.-F., Laurent E., Drusch S., Farruggia A., Durant D. (2024) Potential health value of alternative plant resources explored as feed for ruminants. *30th General Meeting of the European Grassland Federation*, Leeuwarden, The Netherlands

Poster presentation and conference paper

Vitra A., Mesbahi G., Dittmann M. T., Steiner A. K., Thorne S., Hesselmann M., Leiber F. (2024) Effects of potential evapotranspiration on condensed tannin and milk production potential in four grassland mixtures. *30th General Meeting of the European Grassland Federation*, Leeuwarden, The Netherlands

Poster presentation and conference paper

Mesbahi G., Berthet M., Jawahir A., Ginane C., Delagarde R., Chargelegue F., Novak N. (2022) Comportement alimentaire de vaches laitières pâturant des arbres fourragers. *26e Rencontres autour des recherches sur les ruminants*, Paris, France

Poster presentation and conference paper

Mesbahi G., Barre P., Delagarde R., Bourgoin F., Perceau R., Novak S. (2022) Dynamic of 16 fodder trees' nutritive values from June to October. *6th European Agroforestry Conference*, Nuoro, Italie

Oral presentation

Mesbahi G., Jawahir A., Berthet M., Ginane C., Delagarde R., Chargelègue F., Novak S. (2022) Rethinking grasslands in 3D: feeding preferences of dairy cows between temperate fodder trees. *29th General Meeting of the European Grassland Federation*, Caen, France

Poster presentation and conference paper

Mesbahi G., Poutaraud A., Bayeur C., Plantureux S. (2022) Taxonomic and functional biodiversity positively influence agronomic characteristics of permanent grassland. *29th General Meeting of the European Grassland Federation*, Caen, France

Oral presentation and conference paper

Mesbahi G., Bayeur C., Plantureux S. (2022) Production and replacement costs of permanent grasslands compete with those of sown grasslands. *29th General Meeting of the European Grassland Federation*, Caen, France

Poster presentation and conference paper

Plantureux S., Mesbahi G., Bayeur C. (2022) Trade-offs between services rendered by the semi-natural grasslands of the Vosges massif (France). *29th General Meeting of the European Grassland Federation*, Caen, France

Poster presentation and conference paper

Berthet M., Mesbahi G., Duvot G., Zuberbühler K., Cäsar C., Bicca-Marques J.C. (2022) A Brazilian population of titi monkeys reduced to 20% after an outbreak of sylvatic yellow fever. *Joint Conference of the European Federation of Primatology & the Gesellschaft für Primatologie*, Arnhem, Pays-Bas

Oral presentation

Mesbahi G., **Bayeur C.**, **Plantureux S.** (2020) Trade-offs between agronomic and ecological characteristics in permanent grasslands. *28th General Meeting of the European Grassland Federation*, en ligne.

Oral presentation and conference paper

Mesbahi G., **Bayeur C.**, **Plantureux S.** (2019) Les typologies agronomiques : des outils efficaces pour évaluer rendements et performances des prairies permanentes. *Journées de Printemps de l'AFPF*, Paris, France.

Oral presentation and conference paper, in Frenche

Mesbahi G., **Bayeur C.**, **Plantureux S.** (2019) Des prairies permanentes diversifiées et performantes. *Journée de l'école doctorale*, Nancy, France.

Poster

Mesbahi G., **Bayeur C.**, **Plantureux S.** (2018) Prediction of grassland values by phytosociological or agronomical approach. *27th General Meeting of the European Grassland Federation*, Cork, Ireland.

Oral presentation and conference paper

Animal communication

Narbona-Sabaté L., **Mesbahi G.**, **Berthet M.** (2020) Urgent information is shared first in titi monkeys' alarm sequences. *Webinar Société Française de Primatologie*, online.

Oral presentation and conference paper

Narbona-Sabaté L., **Cäsar C.**, **Mesbahi G.**, **Berthet M.** (2020) Black-fronted titi monkeys (*Callicebus nigrifrons*) refine their alarm message after urgent information is shared. *Animal Behaviour Live: Annual Online Conference*, en ligne.

Poster presentation

Berthet M., **Mesbahi G.**, **Pajot A.**, **Cäsar C.**, **Neumann C.**, **Zuberbühler K**. (2018) Titi monkey alarm sequences: when combining creates meaning, in LaCroix T. (Chair), "Evolutionary explanations of compositional communication: the state of the art" symposium. *26th Philosophy of Science Association Biennial Meeting*, Seattle, USA.

Oral presentation

Berthet M., **Neumann C.**, **Mesbahi G.**, **Zuberbühler K.** (2017) Contextual information in titi monkey B-sequences. *7th European Federation for Primatology meeting*, Strasbourg, France.

Oral presentation

Berthet M., **Neumann C.**, **Mesbahi G.**, **Zuberbühler K.** (2017) Do titi monkeys encode predator type and location in their alarm sequences?. *35th International Ethological Conference*, Estoril, Portugal.

Oral presentation

Berthet M., **Neumann C.**, **Mesbahi G.**, **Cäsar C.**, **Zuberbühler K.** (2017) Contextual differences of two alarm call types in titi monkeys. *15th Conference of the Gesellschaft für Primatologie*, Zürich, Suisse.

Poster presentation

Invited speaker

Mesbahi G. (2022) OasYs, Un système laitier agroécologique adapté au changement climatique, *Mise en place et exploitation de haies fourragères*, Grangeneuve, Switzerland

Mesbahi G. (2020) Quelle classification pour la prédiction des propriétés prairiales ? *Séminaire national Habitats - « Typologies, données et cartographies : état des lieux des connaissances et perspectives sur les besoins opérationnels »*, Paris, France

Teaching

2019

Practical — **Agroforestry botany and ecosystem services.**
Master level, 8 hours.
University of Lorraine (France), with PhD Alice Michelot-Antalik

Tutorial — **Statistical analyses of agroforestry dataset.**
Master level, 4 hours.
University of Lorraine (France)

Practical — **Grassland botany and functional traits.**
Master level, 8 hours.
University of Lorraine (France), with PhD Alice Michelot-Antalik

Tutorial — **Statistics: determinant of grassland species and functional traits.**
Master level, 4 hours.
University of Lorraine (France)

2018

Practical — **Collect of animal behaviour on the field.**
Master level, 8 heures
University of Neuchâtel (Switzerland), with PhD Berthet Mélissa

2017

Lecture — **Initiation to global agroforestry**.
Master level, 3 hours.
University of Lorraine

Lecture — **Discovery of research projects**.
Master level, 2 hour.
University of Lorraine, with PhD Jérémie Goulnik

Tutorial — **Mixing crops at different scales.**
Master level, 15 hours. Grading of final exam.
University of Lorraine, with Prof Sylvain Plantureux

Practical **Grassland botany and functional traits.**
Master level, 8 hours.
University of Lorraine, with PhD Jérémie Goulnik

Professional services

Member of the EURAF 2026 Organizing Committee

Workshop co-organization *Les prairies permanentes, notre avenir en herbe !* (2021) Meisenthal, France
120 participants, 20 speakers

Reviewer for *Agriculture, Ecosystems & Environment* (2025)

Reviewer for *General Meeting of the European Grassland Federation* (2020), *IFOAM Animal Husbandry Alliance Conference* (2026)

On-site presentations of the FiBL agroforestry system. For researchers, agricultural advisers, farmers and students (since 2025)

On-site and online presentations of the OasYs farm and its agroforestry system. For researchers, agricultural advisers, farmers and students (2020-2022)
10 presentations

Jury member for *Concours des Pratiques Agro-Ecologiques Prairies & Parcours* (2019)

Outreach

Berthet M. & Mesbahi G. (2025) Os Macacos Sauás
Santuario do Caraça

Interpretive signs, poster and social media posts

RMT AgroforesterieS - Elevage (2025) L'arbre et l'elevage
Genial.ly interactive tree

Online app

Mesbahi G. (2022) Climat, biodiversité : le retour gagnant des arbres champêtres. *The Conversation France*

Article

Berthet M. & Mesbahi G. (2022) Fièvre jaune : les singes aussi se cachent pour mourir. *The The Conversation France*

Article

Mesbahi G. (2021) RadioCocci n°6 Agriculture et changement climatique, online

Radio show

Mesbahi G., **Bayeur C.**, **Plantureux S.** (2021) La typologie agroécologique des prairies permanentes. *Les prairies permanentes, notre avenir en herbe !*, Meisenthal, France

Oral presentation

Mesbahi G., **Novak S.** (2021) L'abre, une ressource fourragère. *Les prairies permanentes, notre avenir en herbe !*, Meisenthal, France

Oral presentation

Berthet M. & Mesbahi G. (2019) Découvrez les singes titis et leur système de communication unique. *The Conversation France*

Article

Mesbahi G. (2018 et 2019) La typologie de prairies permanentes : un outil pour agronomes et écologues. *Journée de formation à la phytosociologie du massif vosgien*, Saverne et Rothau, France

Oral presentation

Mesbahi G. (2018) Les typologies agroécologiques. *Natura 2000 biogeographic seminar*, Parc Naturel Régional des Vosges du Nord, France

Oral presentation

Mesbahi G. (2018) Prairies permanentes : la conservation par la valorisation : exemple du massif vosgien. *Séminaire des Parc Naturels Régionaux Biodiversité et Gestion de l'Espace*, Eppe-Sauvage, France

Oral presentation

Languages

French	Mother tongue	**German**	Basic
English	Fluent	**Portuguese**	Basic